Limit order book spreads, depth, and market efficiency in a general equilibrium model

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ABSTRACT

This paper studies equilibrium order book formation in a limit-order market by building a search-theoretic model driven by expectations about the uncertain arrival of participants. The equilibrium order book's properties, including spread and price impact, are characterized in terms of competitive balance, which endogenously splits the gains from trade between buyers and sellers. We characterize the determinants of welfare, which does not always increase with greater liquidity. We find that the spread is increasing in the gap between sellers' and buyers' valuations, and increases as the competitive balance shifts towards sellers. Additionally, the paper demonstrates how to extract the model's parameters from order book data, applying these techniques to data from Coinbase's Bitcoin/U.S. Dollar exchange. We estimate that, on average, the exchange generates 88% of the welfare that would be possible in a frictionless market.

Keywords: Order book, spread, price impact, equilibrium search, liquidity, market microstructure, welfare, Bitcoin

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1 Introduction

In most equity markets throughout the world, market makers provide liquidity to those wishing to purchase or sell assets by placing limit orders. These limit orders remain on the book until either executed or canceled. An execution occurs when a participant places a market order, which transacts at the most favorable price on the order book.

This arrangement leaves each side of the market with distinct uncertainty. Those who place limit orders choose the price at which their orders will transact, but must await the arrival of enough market orders to reach that price. Meanwhile, those who place market orders are expressing a desire to transact immediately; yet they face uncertainty in the transaction price if several market orders occur in close succession. Moreover, changing market conditions could alter either side's motivation to trade. Understanding how these sources of uncertainty change the incentives of liquidity providers and liquidity demanders is essential to understanding the efficiency of these exchanges as well as the trading opportunities that they are likely to provide.

This paper presents a model of order book trading where these uncertainties are all rooted in the random arrival rate of liquidity providers (those placing limit orders) and liquidity demanders (those placing market orders). For ease of exposition, we focus on the ask side of the market and refer to liquidity providers as sellers and liquidity demanders as buyers. The bid side of the market would proceed similarly. In this model, sellers understand that a higher price will only transact when buyers are especially plentiful, and buyers anticipate that their transaction price depends on how many other buyers arrive near the same time. We depict their decision making in a search environment.[1]

Traders enter the model knowing their valuation of the asset. Sellers share a common view of the asset's fundamental value, which is strictly less than the valuation of all buyers.[2] Thus, trade is always beneficial, which the order book facilitates by determining the terms of trade. Both sides of the market may postpone participation and not place an order if it is optimal to abstain. Despite the homogeneity of participants, the order book always generates a range of prices, without requiring the presence of noise traders who would transact regardless of the prices they face. This setup allows us to study several important equilibrium properties

[1] The presence of an order book means search is not random — participants know where to find partners who would benefit from trade — but there is uncertainty about how many partners will be there. The order book also differs from directed search (as surveyed in Wright et al., 2021), in that all participants interact in a single market with transaction prices depending on realized market tightness (or competitive balance as defined in this paper), rather than offering differing price/market tightness pairs across submarkets, as in directed search models.

[2] That is, we explore the extent to which order book prices can be explained from variation in participant arrivals rather than variation in participant valuations.

at one time, including the spread, the price impact of trades of any size and the welfare available from trade.

This paper demonstrates that the net effect of participating traders' behavior on the equilibrium order book can be characterized in terms of *competitive balance*, which measures the fraction of surplus that a seller receives from the average transaction. This metric is derived from the joint distribution of buyer and seller arrivals. While one might expect that the competitive balance always shifts towards buyers as sellers are more plentiful, this is not always the case. In particular, buyers are better off when the arrival rates are more similar on both sides, while sellers benefit when it is more likely that there are fewer sellers than buyers.

This relates to the broader concept of *market liquidity* that is used in practice to characterize how easily one can transact in a limit order book market. Greater liquidity manifests in a narrower spread (*i.e.* the difference between the highest bid limit order and the lowest ask limit order) or in a smaller price impact (*i.e.* if buying X shares, the difference between the first share's price and the X^{th} share's price). In the literature, there is evidence that a lack of liquidity is perceived by traders to be risky in the aggregate (Pástor and Stambaugh, 2003), causes higher fees associated with seasoned equity offerings (Butler et al., 2005), and leads to less informative prices (Kerr et al., 2020). In our model, the market becomes more liquid as the competitive balance shifts towards buyers. Thus, more liquid markets are better for buyers, but worse for sellers.

Since we fully model trader preferences, we can evaluate the expected welfare associated with the market, and find that greater liquidity does not necessarily imply greater welfare. In some cases, welfare and liquidity move in tandem (such as when market participants have greater parity in arrival rates), but these concepts can also diverge. For example, as the joint surplus of each transaction increases, expected welfare, spread, and price impact all increase—thus increasing welfare, while decreasing liquidity. Moreover, a shift in competitive balance away from sellers will always improve liquidity, but can eventually reduce welfare if it leads to many unmatched sellers. These welfare consequences are presented theoretically as well as quantitatively for the sample of data that we analyze.

To illustrate the model's explanatory power, we apply it to order book data from Coinbase's Bitcoin/U.S. Dollar exchange.[3] Our calibration process interprets the data through the lens of our model. First, we parse the flow of updates to the order book into time periods (regimes) in which the best ask and best bid prices do not cross during the regime. We apply our model to each regime separately by estimating the average order book within

[3] These data are convenient to collect and of independent interest, but the model is equally applicable to any limit-order market.

each regime. We leverage the structure of the model (using the equal profitability of various limit orders) to determine implied beliefs about the arrival of market participants. Doing this sheds light on the underlying forces that drive liquidity and welfare, and allows us to estimate both welfare in this exchange for the period in question.

We find that the average estimated asset valuation difference between buyers and sellers is \$0.99 in these data, with an observed average spread of \$0.16. The competitive balance lies heavily in favor of buyers, as we find that just under 14% of the gains from trade go to the sellers on average. We document the role that implied beliefs about the distribution of trader arrival and differences in trader valuations have on the competitive balance between buyers and sellers, and study the extent to which these predict the observed spread. We find that in this sample approximately 30% of the variation in the spread can be explained by differences in competitive balance, while the remainder is due to variation in the difference between buyer and seller valuations. The market is reasonably efficient, delivering on average 88% of the welfare that could be obtained from a frictionless market. We also examine what drives the variation in welfare across regimes and decompose this into its component parts, the difference in trader valuations being the most important determinant.

We see the contribution of this paper as three-fold. The first contribution is that we provide an equilibrium model that makes predictions about both the spread and the price impact in a limit order market. Our analysis is built on the uncertain arrival of participants in the market, which is the fundamental motivation for organizing a market with an order book. Second, our paper allows us to analyze the welfare implications of order books because all traders are modeled explicitly from their preferences. Finally, our paper makes a methodological contribution by demonstrating how to extract beliefs about arrival of participants from typical order book data, and then use these beliefs to evaluate expected welfare.

We proceed in the paper by first reviewing the literature, after which we build a general model in section 2, analyzing competitive balance, liquidity and welfare in equilibrium. In section 3, we illustrate how to apply the general model to observable order book data, assess the fit of the model, and evaluate the welfare in this market. Section 4 concludes.

1.1 Literature

The literature on order book formation begins with the canonical models of Glosten and Milgrom (1985) and Kyle (1985).[4] Both models include a fraction of uninformed noise traders, who must purchase regardless of the market price. Market makers do not know whether market orders come from an informed or uninformed trader; so they react to this

[4]Back and Baruch (2004) nests both models in a continuous time model and establishes conditions under which equilibrium converges to that of Glosten and Milgrom (1985) and Kyle (1985).

potential adverse selection by updating their beliefs about the asset's value in the market by observing the order flow. They find that when competition increases among market makers, the spread tightens (in Glosten and Milgrom, 1985) or the price impact flattens (in Kyle, 1985). These are both true in our setting, though we do so without the presence of noise traders, which enables us to examine welfare consequences for the market as a whole. Easley and O'Hara (1987) develop the implications of adverse selection further, building a model where larger orders are more likely to come from an informed seller.

Glosten (1994) features similar adverse selection, but rather than sellers learning dynamically from individual transactions, the order book reflects a static, cross-sectional snapshot of expectations, indicating the expected marginal valuation if orders reach a given level. Biais et al. (2000) introduces imperfect competition into this setting, with a finite set of market makers that provide less than the efficient level of liquidity in equilibrium. The cross-sectional snapshot in both papers is similar to the approach in our model. Our setting has a continuum of sellers, but rather than reaching perfect competition, sellers exercise some market power due to the fact that buyers may not transact.[5]

An order book can organize market interactions even when information is symmetric. In Hollifield et al. (2004), trade is motivated by idiosyncratic differences in private valuations, so sellers who value their asset more submit higher limit orders, and are willing to accept a lower (exogenously specified) probability of selling it. Our model eliminates uncertainty about valuations entirely, but the uncertain realizations of buyer/seller ratios endogenously generate an order book that balances the probabilities and prices of trades. Hollifield et al. (2004) and its extension to continuous time in Hollifield et al. (2006) apply their models to order book data with Ericsson stock data from the Stockholm Automated Exchange in the former and with three mining stocks traded on the Vancouver Stock Exchange in the latter. The latter also evaluates market efficiency, finding that the market realizes 90% of the potential gains from trade from a perfectly liquid market, with inefficiency occurring due to non-execution of some limit orders. We find similar quantitative welfare results, although our results are driven by the mismatch in the arrival of participants that prevent some transactions from occurring while they are still socially beneficial.

Another strand of literature has modeled centralized exchange in the presence of search frictions.[6] These build dynamic models of limit-order markets where all traders' preferences

[5]Note that Glosten (1994) examines competition across exchanges, which we do not. All five of these adverse selection papers focus on theory, not applying the model to data as we do.

[6]Search theory is frequently applied in studying over-the-counter exchange, where buyers and sellers must find each other through a decentralized process. Weill (2020) provides a recent survey, including the relatively sparse application to centralized trade. Beyond asset markets, Coles and Smith (1998) examine marketplaces more generally, where the new buyer or seller either matches with the existing stock upon entry (similar to a market order), or if not, becomes part of the stock and awaits a match with the flow of new entrants (similar

are fully modeled, as in our paper. In Goettler et al. (2005), traders can only cancel old limit orders with some exogenous probability. All traders agree on the value of the asset at a given time, but that value can change each period, creating a risk that a limit order is executed when it is no longer beneficial to the trader who offered it. Trade in Foucault et al. (2005) is motivated by differing waiting costs, where low cost traders post limit orders that are executed by market orders from high cost traders. By assumption, new limit orders must be price improving (i.e. narrow the spread) by the exogenous tick amount. Roşu (2009) has a similar setting except that neither a minimum tick nor price improving requirement is imposed; rather, both arise endogenously as the most profitable limit order to add to the order book. Dugast (2018) focuses on changes in investor valuations over time, generating predictions about the relationship between the frequency of news arrival, market depth, and executions.

For each of the papers in the previous paragraph, the order book is dynamically built up over time as arriving traders either place limit orders or remove liquidity by placing market orders. This generates predictions about low-level trends from specific transactions on the order book: the correlation between transaction costs and spread, the distribution of market spread and execution times, and bid and ask prices display co-movement, respectively. In contrast, our model characterizes the order book averaged over short periods where asset fundamentals are steady but arrival of traders may vary. This allows us to examine a more aggregate perspective: the distribution of limit orders informs us of the anticipated arrival rates of participants and thus reveals expected welfare. Our model also allows free entry and exit from limit orders, and approximates a large market with a continuum of traders. This generates an analytic solution that is particularly useful in deriving empirical implications for the order book that we can take to data.

Several search theoretic papers study liquidity in asset markets from a macro perspective. Lagos and Rocheteau (2009) studies liquidity in over-the-counter markets. This model makes predictions about liquidity measures like the bid-ask spread as in our paper, although they do not attempt to model the mechanics of price priority in limit-order books as is done here. Cui and Radde (2016) also abstracts from the specific properties of the limit-order book, embedding a financial sector with search frictions into a dynamic general equilibrium consumption-saving-investment model. Vayanos and Wang (2007) builds a search-based model of trading with the friction that traders can search for only one asset and then studies the equilibria that result. They show that, in one equilibrium, short-horizon investors

to placing a limit order). Matching there is a matter of finding an acceptable good, rather than a mutually beneficial price. Ebrahimy and Shimer (2010) applies a similar organization to labor markets, where firms with a new job posting inspect workers in order, starting with those unemployed the longest.

congregate in one market which is more liquid than the other market with long-horizon investors. The question of time horizon does not enter into our model since all traders have the same trading horizon.

2 An order book model

Consider a market for an asset, with risk neutral sellers and buyers. Sellers are homogenous, each holding one unit of the asset from which they obtain value d_A.[7] Buyers are also homogenous, with demand for one unit of the asset that they value at $d_B > d_A$.

In the model, a random number of buyers and sellers enter the market. Upon entry but before knowing the realized number of participants, each seller simultaneously places a limit order, which means he selects a price at which he is willing to sell. The set of limit orders form the order book. Then, each buyer simultaneously places a market order, which means she will transact at the lowest available price on the order book, though the buyer can abstain if available prices are too high.[8]

A buyer in a limit order market does not know whether other traders have submitted orders that will be executed in the time between when she submits her order and when it is received by the exchange. We capture this real-world dynamic by placing the buyers in random order, then transacting in sequence from the lowest order book price upward. When transactions are realized, the exchange pays the seller a rebate f_m and collects a take fee f_t from the buyer. We take these exchange fees to be exogenous.[9] After all possible transactions are finished, either because no more buyers remain or because no more acceptable seller prices remain, all participants exit the market.

Given an order book, the realized price for any submitted market order depends on the ratio of submitted buy orders to the number of sell orders. We denote the realized ratio of buyers to sellers as q, which is a random variable drawn from cumulative distribution $F(q)$ for $q \geq 0$. We assume the random variable q to be continuously differentiable for all q, with $F'(q) > 0$ for all $q \leq 1$. This realization of q is the source of uncertainty for both sides of the market in terms of whether a transaction will occur and at what price. If realized q is large

[7] Incorporating multi-unit supply or demand would only be consequential in our model if a single agent's order would significantly alter the order book. In that case, the agent (buyer or seller) may want to strategically spread out the order over time.

[8] This behavior is consistent with buyers' submission of marketable limit orders—that is, limit buy orders that have a price that crosses the spread. Throughout the paper we use the term "market orders" as a shorthand for these marketable limit orders.

[9] In practice, the rebate is meant to encourage sellers to provide liquidity, while the take fee can pay for the rebate plus any cost of operating the exchange. One could endogenize the fees by modeling the exchange and its competitive environment, but such a model would require this current model as an input and the focus of this paper is understanding the shape of the limit order book and its welfare implications.

then there are many buyers relative to sellers. As we will see, this distribution determines the equilibrium order book and the expected welfare in the market.

2.1 Sellers

Each seller receives value d_A while holding the asset. The seller selects an ask price a for his limit order, knowing that the order will only transact if a becomes the lowest price on the order book (*e.g.* all lower prices have transacted). This occurs with equilibrium probability $G(a)$, which the individual seller takes as given, though it will be endogenously determined. Upon execution, the seller is paid a by the buyer as well as the rebate f_m by the exchange, and exits the market. If the limit order fails to transact, the seller retains the asset and receives utility d_A. Thus, the expected utility for a seller entering the market, V_S, is computed as:

$$V_S = \max_a G(a)(a + f_m) + (1 - G(a))d_A. \tag{1}$$

2.2 Population Distribution

The ratio of buyers to sellers q is independently drawn from the exogenous distribution $F(q)$, which represents the exogenous beliefs of agents on both sides about the relative number of shares that will be desired by market orders relative to shares offered as limit orders. This model does not inspect the rationality of those beliefs; rather, we will use order book data to back out what market makers must believe about those arrival rates.

As an example of such a distribution,[10] suppose that the number of shares n_a that sellers offer as limit orders are exponentially distributed with mean $1/\alpha$, while the number of shares n_b that buyers seek as market orders are independently exponentially distributed with mean $1/\beta$. The joint distribution then becomes $f(n_a, n_b) = \alpha e^{-\alpha n_a} \beta e^{-\beta n_b}$. If we define the ratio of market orders to limit orders as $q = \frac{n_b}{n_a}$ and define $\phi = \frac{\beta}{\alpha}$ (the average number of sellers per buyer), this cumulative density can be transformed to become $F(q) = \frac{\phi q}{1+\phi q}$.

If the realized ratio q is less than 1, all market orders to buy are executed but some limit orders to sell will not be. Indeed, the realized ratio of buyers to sellers q will dictate how much of the order book is used up, and therefore implies what the lowest remaining price $a(q)$ must be: namely, the endogenous probability of reaching price $a(q)$ must equal the exogenous probability of having drawn ratio q or higher.

$$G(a(q)) = 1 - F(q) \text{ if } q \leq 1. \tag{2}$$

[10] Our results do not depend on the functional form in this example, but we will refer to it where useful for intuition.

This expression encodes one of the key properties of limit order book markets: liquidity takers are always given price priority in their transactions. The lowest prices transact first, so a seller who lowers his price $a(q)$ increases the probability of selling, even while giving up some revenue.

Note that if the realized ratio $q > 1$, then all of the limit orders on the book that are compelling to buyers are executed, leaving some market orders that will not be executed. Thus, $q < 1$ results in some sellers not transacting, while $q > 1$ results in some buyers not transacting.

2.3 Buyers

A buyer submitting a marketable limit order in this market cannot control the price she receives, which depends on the order in which buyers are served, but can set a maximum price $\bar{a}$ beyond which she prefers to not transact. Like sellers, buyers are assumed to know the distribution $F(q)$, though not the realization of q. If $q \leq 1$, all buyers are able to transact, with a buyer's position in the queue s being uniformly distributed over $[0, q]$. If $q > 1$, the sales will be rationed among buyers, serving only $1/q$ of them, each of whose queue position s is uniformly distributed over $[0, 1]$. If the buyer transacts, she pays the take fee f_t and the realized price $a(s)$ and receives utility d_B. That is, the buyer is equally likely to be in any position in the order arrival process, and that order determines which order-book price the buyer receives. Those who are unable to transact receive 0.

Thus, the expected utility for a buyer entering the market, V_B, is computed as:

$$\begin{aligned} V_B = &\int_0^1 \left(\int_0^q \frac{1}{q} \left(\max\left\{ d_B - a(s) - f_t, 0 \right\} \right) ds \right) F'(q) dq \\ &+ \int_1^\infty \left(\int_0^1 \left(\max\left\{ d_B - a(s) - f_t, 0 \right\} \right) ds \right) \frac{1}{q} F'(q) dq. \end{aligned} \tag{3}$$

The first term in parentheses handles realizations of $q \leq 1$. In this case the buyer does transact, and their realized position s in the order queue is distributed with density $1/q$, determining the price $a(s)$ of that transaction. The second term handles realizations when $q > 1$, so only fraction $1/q$ of buyers transact, while the trader's realized position in the order queue s is distributed with density 1.

In this setting, buyers receive the best price available on the order book when their order hits the market, so their only strategic choice the setting of the maximum price at which they will transact. This results in a reservation price $\bar{a} = d_B - f_t$.

2.4 Equilibrium

Liquidity providers make strategic decisions simultaneously, creating a normal form game. The Nash equilibrium consists of a pricing function $a(q)$ for $q \leq 1$, transaction probabilities $G(a)$, and a reservation price $\bar{a}$ such that:

1. The support of $G(a)$ equals $[a(0), a(1)]$.
2. Every price in the support of $G(a)$ is equally profitable, maximizing Eq. 1, while prices outside the support are weakly less profitable.
3. The order book $a(q)$ anticipates the expected population distribution in Eq. 2
4. Buyers are willing to accept all prices in the support, meaning $\bar{a} = a(1)$ is the optimal reservation price for Eq. 3.

Conditions 1 and 4 ensure that all prices offered on the order book can be executed under some realization of q. If Condition 4 failed, sellers would be listing some prices that no one would ever accept, violating Condition 2. Condition 2 adds that sellers must be indifferent across all the prices on the order book. If one were strictly more profitable (in or out of the support), all sellers would deviate to offer only that price. Condition 3 ensures that prices are consistent with agents' expectations about the possible ratios of buyers to sellers.

2.5 Equilibrium solution

We proceed by presenting a candidate equilibrium, then prove that it uniquely satisfies the equilibrium requirements.

As a preliminary, we introduce some notation used throughout the equilibrium solution and the calculations of liquidity and welfare. First, we define X_t as the probability that a buyer (i.e. liquidity taker) successfully transacts in a given attempt, as derived from the exogenous distribution of buyer/seller ratios:

$$X_t \equiv F(1) + \int_1^\infty \frac{1}{q} F'(q) dq. \tag{4}$$

Note that with probability $F(1)$, all buyers transact, but when $q > 1$, each buyer transacts with probability $\frac{1}{q}$.

When the buyer does transact, her order could be the first to arrive and pay price $a(0)$, or could be the last and pay $a(q)$. As a step in computing the expected price they will pay, we define X_p as follows:

$$X_p \equiv (1 - F(1)) \int_0^1 \left(\int_s^\infty \frac{F'(q)}{q} dq \right) \frac{1}{1 - F(s)} ds \tag{5}$$

This generates a weighted average of the position of the successful buyer s across all realizations q where $q > s$. The weighting $\frac{1}{1-F(s)}$ is relevant for how much the price increases with position s, as we show below.

For the exponential distribution of agents introduced in Section 2.2, these metrics evaluate to $X_t = \phi \log\left(1 + \frac{1}{\phi}\right)$ and $X_p = \frac{1}{2}\left(\log\left((1+\phi)\left(\frac{1+\phi}{\phi}\right)^{\phi^2}\right) - \phi\right)$.

Finally, we define Δ as the net gain when a transaction occurs:

$$\Delta \equiv d_B - d_A - f_t + f_m \tag{6}$$

The first term comes from the transfer of the asset to the buyer who values it more. This is reduced by the net exchange fees.[11] We assume throughout that the net gain Δ is strictly positive.

With this notation in place, we can then provide the equilibrium solution. Prices have the following function of the buyer/seller ratio q:

$$a(q) = d_A - f_m + \Delta \cdot \frac{1 - F(1)}{1 - F(q)}. \tag{7}$$

The probability that a seller transacts when listing price a is:

$$G(a) = \frac{\Delta \cdot (1 - F(1))}{a - d_A + f_m}. \tag{8}$$

This distribution has a support of $[a(0), a(1)]$, where:

$$\begin{aligned} a(0) &= d_A - f_m + \Delta(1 - F(1)) && (9) \\ a(1) &= d_B - f_t. && (10) \end{aligned}$$

Proposition 1. *Equations 7 through 10 constitute the unique equilibrium in this market.*

The proof is presented in the appendix. We show by construction that the proposal is an equilibrium, and then establish uniqueness by showing that any other price distribution would result in profitable deviations. For instance, a collusive order book in which all sellers

[11] For instance, $f_t - f_m$ would be the exchange revenue. If this is used to exactly cover exchange cost, then the net fees indicate resources lost in facilitating the transaction.

list the same high price is not an equilibrium, because a seller who undercuts this price by ϵ would strictly increase the probability of transacting and thus increase profits.

2.6 Competition and Limit Order Pricing

The equilibrium solution reveals how the gains from trade are split between buyers and sellers in this order book market. In the absence of trade, a seller would retain the asset and enjoy utility d_A. When placing a limit order in this market, the seller's expected utility (as shown in the proof of Proposition 1) is:

$$V_S = d_A + \Delta \cdot (1 - F(1)). \tag{11}$$

Recall that Δ is the social gain from trade (moving the asset to a person who values it more), net of exchange fees, while $1 - F(1)$ is the probability that at least some of the buyers will not find a seller. Thus, in equilibrium, $1 - F(1)$ also indicates the expected fraction of social gains that accrue to sellers.

As $F(1)$ increases, it becomes more likely that some sellers do not transact. This intensifies competition among them, shifting the limit order book towards lower prices. Indeed, if $F(1) = 1$, sellers reach Bertrand competition, with everyone offering their marginal cost of $d_A - f_m$. Thus, the quantity $1 - F(1)$ characterizes the fraction of the potential gains to trade Δ that are captured by sellers. In light of this, we will often refer to the fraction $1 - F(1)$ as the competitive balance. This fraction plays an important role in both market liquidity and welfare.

In the absence of trade, a buyer would not obtain the asset and have 0 expected utility. In a limit order market, while V_B is positive, the frictions of the market mean that buyers and sellers do not just split the total potential gains from trade, Δ. The equilibrium expected utility of buyers (as shown in the proof of Proposition 1) also depends on the the frictions inherent in order book markets and is given by

$$V_B = \Delta \cdot (X_t - X_p). \tag{12}$$

Note that $X_t - X_p < F(1)$, which implies that for a buyer and seller that will eventually match, the expected gains from trade are

$$\Delta \cdot (X_t - X_p + 1 - F(1)) < \Delta, \tag{13}$$

meaning that some of the potential benefits of trade are dissipated relative to frictionless trade. This is because the order book mechanism cannot ensure that everyone will be served at a given time. This inefficiency arises from a (probabilistic) imbalance in the number of buyers who arrive and sellers who arrive and will be discussed in the welfare section.

With the crucial role of $F(1)$ in market outcomes, it is worth exploring what information it encapsulates. First, consider an extreme case where buyers are almost always more abundant than sellers; that is, let $F(1) = 0$ and let $\chi \equiv \int_1^\infty \frac{1}{q} F'(q)dq$. This results in $X_t = \chi$, while $X_p = 0$ (because $F(s) = 0$ for all $s \leq 1$). Indeed, in this scenario, sellers extract all surplus from buyers, because of their relative scarcity: $V_S = d_A + \Delta$ and $V_B = 0$.

In the opposite case, consider what happens when the q for which $F(q) = 1$ approaches 0. This means that sellers are increasingly abundant relative to buyers and $X_t \to 1$, while $X_p \to 0$. In this case buyers extract all of the available welfare in the limit; that is, $V_S = d_A$ and $V_B = \Delta$.

For less extreme scenarios, we consider how changes in the anticipated balance of buyers and sellers in the market will affect the market participants. Consider the effect of compressing the distribution F around $q = 1$. This result is meant to aid in understanding how increasing the likelihood that both sellers and buyers are able to transact changes welfare. It shows that $X_t - X_p$ falls when a distribution places more weight (in a first-order stochastic dominance sense) near $q = 1$ (indicating more balance in the arrival of buyers and sellers). We show this in two parts, comparing the effect from the range where $q > 1$ and then the range where $q < 1$.

Proposition 2. *Consider two distributions of participants, F and $\hat{F}$. Let V_S and V_B be the expected utility under the F distribution, and similarly $\hat{V}_S$ and $\hat{V}_B$ under the $\hat{F}$ distribution. Then $\hat{V}_B > V_B$ if:*

1. *$F(q) = \hat{F}(q)$ for all $q \leq 1$, and $F(q) < \hat{F}(q)$ for all $q > 1$, or*
2. *$F(q) > \hat{F}(q)$ for all $q < 1$, and $F(q) = \hat{F}(q)$ for all $q \geq 1$.*

Moreover, $\hat{V}_S = V_S$ if $\hat{F}(1) = F(1)$ and $\hat{V}_S > V_S$ if $\hat{F}(1) < F(1)$.

Thus, the expected utility of buyers increase when there is more balance between buyers and sellers. This is to be expected in the first case, since fewer buyers are left unserved in expectation. Surprisingly, in the second case, buyers are better off when fewer *sellers* are left unserved as well. The higher likelihood to sell induces competition among sellers that reduces their limit order prices. The expected utility of sellers, on the other hand, only depend on the probability that they collectively sell out, not the rest of the distribution of arrivals. If $F(1)$ is unchanged (as it is in both cases of Proposition 2), their utility is constant.

Of course, Proposition 2 only applies to changes in $F(q)$ that shift weight towards $q = 1$. If the distribution of participants shifts weight towards 1 in some regions and away from 1 in other regions, the impact on buyer welfare would be ambiguous. For example, in our exponential example, an increase in ϕ will increase $F(q)$ at each q, which mixes the first part of the proposition with the opposite of the second part. In that example, the competing effects net out such that the competitive balance $1-F(1)$ is strictly decreasing in ϕ. That is, the first effect dominates, benefiting buyers as fewer of them are left unserved and harming sellers.

2.7 Expected Welfare

We next consider the expected welfare of all market participants in an order book market. First, we establish two extremes—one with no trade and the other with frictionless trade. Autarky would leave the asset with sellers indefinitely, generating a total welfare of d_A for the seller. At the other extreme, if there were no frictions from the timing of trades, each seller who enters the market would immediately find a buyer, generating for them a welfare of $a + f_m - d_A$, where a is the price at which they sell the asset. The buyer's welfare if they were certain to find a seller when they enter the market is $d_B + f_t - a$. Thus, the total welfare for the buyer and the seller in this transaction would be $a + f_m - d_A + d_B + f_t - a$, which we have already defined as Δ.

We compare this frictionless benchmark to the gains in an order book market. If we always had equal flows of buyers and sellers, the market would generate V_B per buyers and V_S per sellers, compared to 0 and d_A in autarky, respectively. Thus the welfare per transaction are: $V_B + V_S - d_A = \Delta \cdot (1 - F(1) + X_t - X_p)$. However, this neglects the potential imbalance in participants: there can only be n_b transactions if there are more sellers than buyers ($n_a > n_b$), and only n_a transactions otherwise. If we rephrase this as welfare per buyer, there will be 1 transaction per buyer if $q < 1$, and $\frac{1}{q}$ transactions per buyer otherwise. We can then average over the possible realizations of q to get expected welfare per buyer:

$$
\begin{aligned}
W &= \Delta \cdot (1 - F(1) + X_t - X_p) \cdot \left(\int_0^1 1 \cdot F'(q) dq + \int_1^\infty \frac{F'(q)}{q} dq \right) && (14) \\
&= \Delta \cdot (1 - F(1) + X_t - X_p) \cdot X_t. && (15)
\end{aligned}
$$

Expected welfare per buyer has three components. The first, Δ, is the first best outcome from frictionless trade, as previously noted. Note that the other two components are always weakly less than 1. The third term, X_t, is the probability that a buyer transacts. This exclusively reflects the potential for mismatch with too few sellers. The second term (in parentheses) reduces the gains from trade because of the potential loss of opportunity to trade. We label this component Γ, and call it the friction component of welfare. We note that the friction component $\Gamma = 1 - F(1) + X_t - X_p$ includes the term $1 - F(1)$ which represents the perceived probability that all sellers will be able to transact. This term will be relevant for the liquidity discussion to come.

Both the second and third terms in the welfare calculation depend on market makers' beliefs about the distribution of the arrival of buyers and sellers. Changes in these beliefs will alter the expected welfare from the operation of the market. We consider how the welfare is affected when the distribution F is compressed around $q = 1$, in parallel to Proposition 2.

Proposition 3. *Consider two distributions of participants, F and $\hat{F}$. Let W and $\hat{W}$ be the expected welfare under the respective distributions. Welfare $\hat{W}$ is higher than W if:*

1. *$F(q) = \hat{F}(q)$ for all $q \leq 1$, and $F(q) \leq \hat{F}(q)$ for all $q > 1$, or*
2. *$F(q) = \hat{F}(q)$ for all $q \geq 1$, and $F(q) \geq \hat{F}(q)$ for all $q < 1$*

Both cases depict the distribution shifting toward greater parity between buyers and sellers, which increases the number of transactions that occur (all of which have a net welfare gain of Δ). This is true even in the second case, where the $\hat{F}$ distribution makes buyers less scarce in the times that they are the short side of the market. This is a surprising result since welfare is calculated on a per-buyer basis, so having more buyers on average risks diluting the gains; but the additional transactions that occur more than compensate for the larger population.

Of course, Proposition 3 does not consider distributional shifts that alter $F(1)$, the likelihood that all buyers are served. Such a change would need to be evaluated with a particular functional form. For our exponential example as previously noted, an increase in ϕ will shift the competitive balance towards buyers, always reducing $1-F(1)$. Even so, this can decrease expected welfare because it is more likely that very few sellers match. We illustrate this in the next subsection.

2.8 Market liquidity and Welfare

In limit-order markets, an asset is typically defined as being more liquid if it has a smaller spread or if a given execution has a smaller price impact.[12] Often, it is taken for granted that greater liquidity is welfare improving, a conjecture that we examine here.

[12]In contrast, Chaderina et al. (2022) finds that during a fire sale, more liquid assets can have a larger price impact because they are commonly held across many institutions, creating a negative externality on other holders.

From Eq. (7), we can calculate the spread and price impact in this market as a function of the model parameters. We define the spread to be

$$S = a(0) - (d_A - f_m) = \Delta \cdot (1 - F(1)), \tag{16}$$

which is the difference between the lowest price that the order book ever offers, $a(0)$, and the price that would come from a competitive market.[13]

The price impact for an order that absorbs the fraction q of the order book in our model is defined to be

$$PI(q) = a(q) - a(0) = \Delta \cdot \frac{(1 - F(1)) \cdot F(q)}{1 - F(q)}. \tag{17}$$

Equations (16) and (17) show that market liquidity as characterized by spread and price impact depend on market characteristics, summarized in the following result.

Proposition 4. *Equilibrium spread and price impact decrease when:*

1. *the competitive balance* $1 - F(1)$ *decreases.*
2. *the difference between buyer and seller valuations* ($d_B - d_A$*, in* Δ) *decreases,*
3. *the fee spread* ($f_m - f_t$ *in* Δ) *decreases,*

This result, in conjunction with propositions 2 and 3, help clarify the relationship between liquidity and market welfare. We see that since liquidity's dependence on Δ, and $F(\cdot)$ is different from welfare's dependence on Δ and $F(\cdot)$, there is space for increased liquidity to *not* lead to increased welfare. In some cases, welfare moves in tandem with market liquidity, but there are notable exceptions as the next result shows.

Corollary 1. *In comparative static analysis of equilibrium,*

1. *A shift in distribution* F *to* $\hat{F}$*, per Propositions 2 and 3, will lead to greater market liquidity and greater welfare.*
2. *A decrease in* Δ *will lead to greater market liquidity but reduced welfare.*

The first result of Proposition 4 simply indicates that spread is proportional to the probability that all sellers transact. If sellers become relatively more plentiful, $1 - F(1)$ falls,

[13] This definition of spread is proportional to the traditional definition of the spread as the difference between the best ask and best bid, but does not require us to model the bid side of the market. Modeling the bid side of the market is analogous to what we have done here, but for expositional ease we forego. If both the bid and the ask side of the market were modeled and the order arrival process was similar to that of the bid side, the spread would be the difference between the lowest ask and the highest bid, which would be twice the spread defined here.

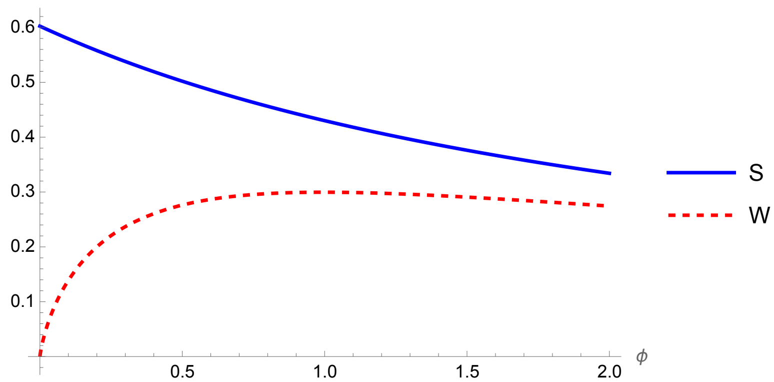


Figure 1: The equilibrium spread and expected welfare as a function of ϕ

as does the spread, indicating greater liquidity. From the results of proposition 2, this would harm sellers yet could help buyers (depending on how X_t and X_p are affected) enough to more than offset harms to sellers, and thus raise total welfare.

The second and third results of Proposition 4 indicate that markets are more liquid as Δ falls. The difference in valuations between buyers and sellers can arise through tailored marketing campaigns, substantive differences in the interpretation of available public data, differences in information gathering technologies or preferences amongst market participants, differences in hedging demands due to alternative portfolio goals, or alternative tax or regulatory constraints among market participants. In situations where this heterogeneity is more extreme, we would expect markets to be less liquid.

Additionally, Δ depends on the fee spread $f_t - f_m$ provided to market participants by the exchange. Exchanges where the fee spread is high will be less liquid, *ceteris paribus*.[14] The increased wedge between seller valuations and buyer valuations that arises from increases in the fee spread leads to less liquid markets.

Whether driven by valuation differences or fee spreads, a decrease in Δ will increase market liquidity, but decrease market welfare! These perceived differences between buyers and sellers are what motivate trading in the first place, so if the net benefit of a given transaction falls, the expected welfare in aggregate falls proportionally. This suggests that one should be cautious about assuming that market liquidity is a clear signal of market welfare.

This distinction between liquidity and welfare can also occur with more general changes in the distribution of participants, F. In our exponential arrival example, as ϕ increases, it

[14] Colliard and Foucault (2012) build a model that focuses specifically on the efficiency of changes in exchange fees. They find that although smaller fee spreads increase gains from trade, when executions happen, they do not necessarily make markets more efficient.

reduces the likelihood of having more buyers than sellers, but this happens both above and below $q = 1$, so the conditions of Propositions 2 and 3 do not apply. As shown in Figure 1, welfare initially increases with ϕ, but hits a maximum (at $\phi = 1.02$ under these parameters) and then is falling thereafter. In contrast, the equilibrium spread is strictly decreasing as ϕ increases. Thus, when buyers frequently outnumber sellers (ϕ is small), greater liquidity also means greater welfare. But when sellers are more plentiful, the spread continues to fall, but this increased liquidity comes with less welfare.

3 Estimating model parameters with order book data

In this section, we study the implications of the model by applying it to order book data from Coinbase's Bitcoin/USD exchange. We first describe key features of the data, then develop a method for extracting analogs of the model parameters from the data. This method includes dividing the data into stable regions called *regimes*, each of which we analyze as instances of the setting to which the model appplies. To demonstrate the insight that the model provides we start by inspecting four sample regimes and describing how the model decomposes liquidity and welfare into its component parts. We then apply this analysis to all of the regimes in the sample and describe the distribution of these welfare and liquidity determinants.

3.1 Data Description

The Coinbase Bitcoin/USD exchange provides a live feed of order book updates through their Websocket API. These updates provide all changes to the order book in near real-time. In order to demonstrate the technique used here, we limit our analysis to one hour,[15] which yielded 1,321,084 order book updates and 14,807 order executions. These updates occur at the rate of approximately 353 updates per second and 3.95 executions per second.[16]

The model, which relies on traders' beliefs about arrival rates and execution probabilities, is most applicable during periods when the order book is stable for long enough that participants can fully adjust to market conditions. In settings with algorithmic traders, that stability can plausibly be achieved in fractions of a second rather than (as in some other financial models) months or years. We define these stable regions as periods where the best bid and ask do not cross. When such a price shift occurs, it sends a clear signal that fundamental parameters have shifted, leading to a new regime.

[15]Data was collected on June 20, 2023, from approximately 22:25:51.04 UTC to 23:28:18.46 UTC.

[16]These data are meant to illustrate the estimation of model parameters using real-world data. It is not intended to be an exhaustive study of the properties of this specific trading venue.

Table 1: Regime summary statistics

Regimes ($N = 2480$)	Duration (sec)	Updates (#)
Mean	1.51	533
St. Dev.	3.07	836
Min	0.00	2
25%	0.04	97
50%	0.35	244
75%	1.60	664
Max	35.93	12820

We use this idea to characterize the regimes to which we apply our model. Specifically, we begin a regime at the start of trading, and track the highest best bid and the lowest best ask over time. We end the regime and start a new one when either the best bid is higher than the initially (at the start of the regime) observed best ask or the best ask is lower than the initially observed best bid. We repeat this process until the end of the sample. In so doing, we generate 2480 regimes. Some of these regimes are very short, lasting only microseconds, while others last for several seconds.

Table 1 gives statistics on the duration and the number of updates for these regimes. Regime duration varies with the median regime lasting about 0.35 seconds. The longest regime lasts about 36 seconds and the mean duration is 1.51 seconds. The median number of order book updates performed during a regime is 244 updates, with a mean of 533 updates and a maximum of 12,820 updates.[17] We anticipate that our model is most applicable to regimes with longer durations; those with extremely short durations occur when both bid and ask prices are rapidly rising or falling, which suggests that the market is still adjusting to changes in fundamental parameters and has not yet reached steady-state. For each regime, the spread is calculated using both sides of the order book by calculating the minimum ask price minus the maximum bid over the regime.

3.2 Estimating $F(q)$ using the order book

To apply the model to the order book data, we need to estimate the distribution of the number of participants in the market, $F(q)$. This distribution is not directly observable, but we can infer it from the observed order book. The model assumes that liquidity providers (market makers) have beliefs about this distribution, and that these beliefs are reflected in

[17]Note that the duration quartiles do not necessarily correspond to the same regime as the corresponding quartile in the distribution of number of updates, since the rate of updates per second is not constant.

Figure 2: Evolution of spread and price impact for regime order books

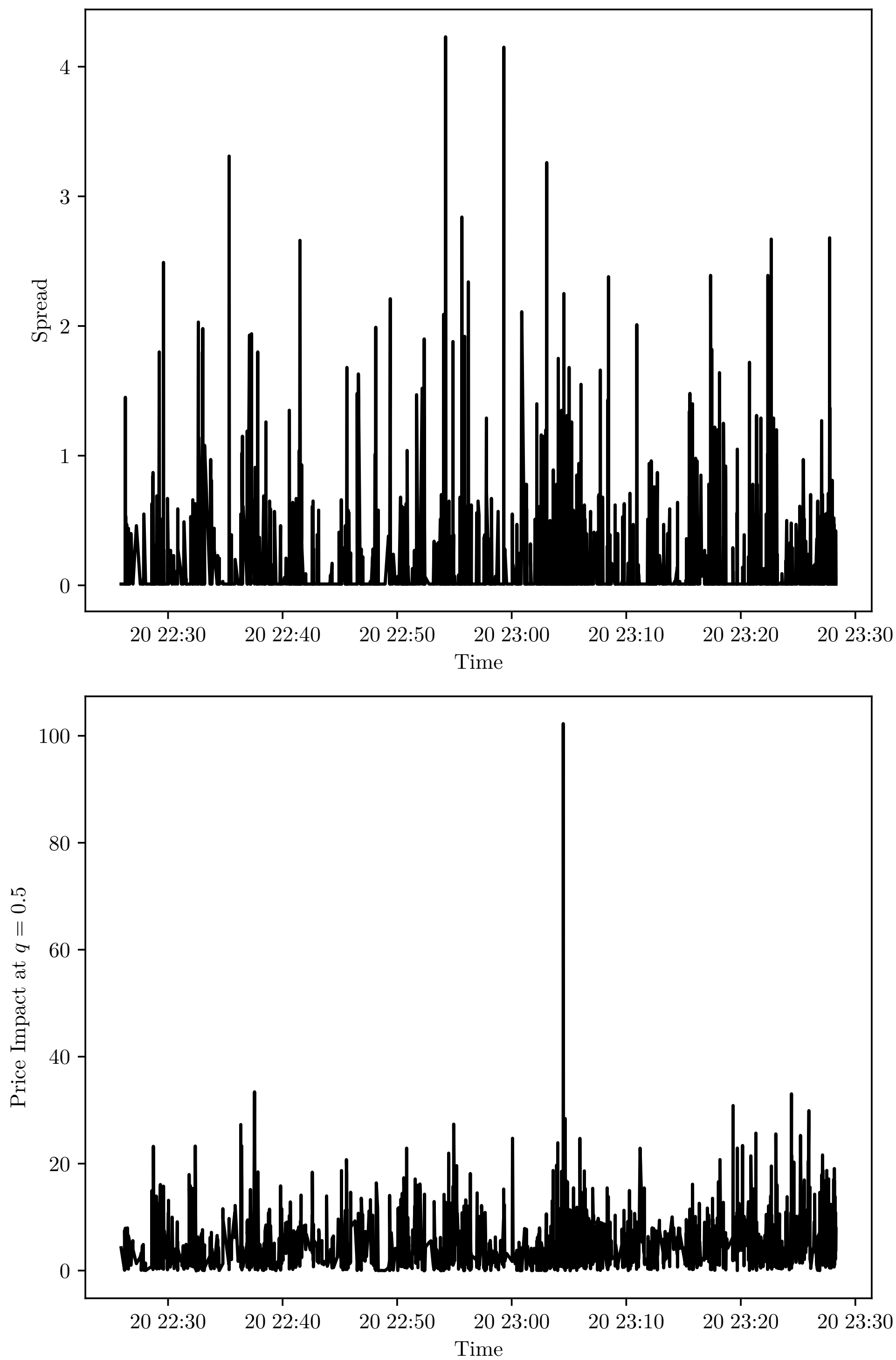


Table 2: Summary statistics of spread and price impact for regime order books

	Spread	Price Impact ($q = 0.5$)
Mean	\$0.16	\$5.10
St. Dev.	\$0.37	\$5.03
25%	\$0.01	\$1.66
50%	\$0.01	\$3.97
75%	\$0.08	\$6.99

their limit orders.

Since the order book allows for limit orders with arbitrarily large price offers, a market maker can costlessly place these even though they are unlikely to ever be executed. Understanding the strategy for placing these "dormant" limit orders is beyond the scope of our model. Instead, we focus on the portion of the order book that regularly executes. We calibrate $q = 1$ to an execution of 1 BTC (covering 99% of executions, per Table 3), and for calculating statistics of the distribution where $q > 1$, we truncate the distribution at $q = 3$ (e.g. 3 BTC). We later demonstrate that our calculations are robust to these choices.

We can then begin to back out the beliefs of liquidity providers about the distribution $F(q)$. This is separately estimated for each regime, using all observations of the order book throughout that regime to generate an average of the underlying beliefs during the regime. This estimation strategy thinks of these regimes as regions within which the information under which traders are participating is stable.

At any time t, we observe a finite set of limit orders that have been placed on the order book. We interpret this as a sample from the distribution of potential seller offers. These sellers know that any price on the order book between $a(0)$ and $a(1)$ is equally profitable. Given their beliefs about $F(q)$ and equations (16) and (17), we see that

$$\frac{PI(q)}{S} = \frac{F(q)}{1 - F(q)}. \tag{18}$$

This relationship allows us to use observations on price impact and spread to back out the implied distribution $F(q)$.

Equation (18) gives an implied CDF of q

$$\hat{F}(q) = \frac{\frac{PI(q)}{S}}{1 + \frac{PI(q)}{S}}. \tag{19}$$

We observe this implied $\hat{F}(q)$ for each order book in the regime. Using these as observa-

Table 3: Percentiles of the distribution of order execution size

Percentile	Execution Size (BTC)
10	0.00025
25	0.00177
50	0.01777
75	0.05318
90	0.17716
99	0.90437

tions on the true CDF $F(q)$, we then fit a mixture of normal CDFs to the data by binning the observed qs into n_b bins where n_b is calculated using Doane's method.[18] Then we divide the data into each bin and select CDFs that have a standard deviation equal to the overall aggregate standard deviation and a mean equal to the mean of samples in that bin. This generates a collection of normal CDFs $\Phi_1, \ldots, \Phi_k$ and the CDF of that regime is taken choosing weights $w_1, \ldots, w_k$ that minimize the sum of squared errors between the observed $\hat{F}(q)$ and the estimated $\overline{F}(q) = w_1\Phi_1(q) + \cdots + w_k\Phi(q)$ subject to the constraint that $\overline{F}(3) = 1$ and that the CDF be monotonically increasing.[19] The estimates of price impact seen previously in Table 2 are calculated using the estimated $\overline{F}(q)$ for each regime.

Given the estimated CDF $\overline{F}(q)$, we calculate $F(1)$, X_p and X_t per Eqs. 4 and 5. After this, we can decompose the spread into the valuation differences between buyers and sellers (Δ) and the measure of seller competitiveness $\Gamma = 1 - F(1)$ using the spread in Eq. (16). Given estimates of $F(q)$, Δ, $1 - F(1)$, X_t, and X_p we can calculate the welfare estimate of that regime using equation (14). In this process, optimization for the mixture weights or the numerical process of estimating the market statistics from these distributions does not converge for 20 of the 2480 (0.81%) regimes. We exclude these regimes from the discussion that follows, meaning the number of regimes for the following analysis is 2460.

[18] In Doane's method (as described in Scott, 2015), the number of bins n_b is given by

$$\begin{aligned} n_b &= 1 + \log_2(N) + \log_2\left(1 + \frac{|g_1|}{\sigma_{g_1}}\right) \\ g_1 &= \overline{\left(\frac{x-\mu}{\sigma}\right)^3} \\ \sigma_{g_1} &= \sqrt{\frac{6(N-2)}{(N+1)(N+3)}}. \end{aligned} \tag{20}$$

[19] This method is inspired by kernel density estimation, but has been adapted because we do not directly observe draws from the density $f(q)$, but instead observe implied values of the CDF $F(q)$.

3.3 Four Case Studies

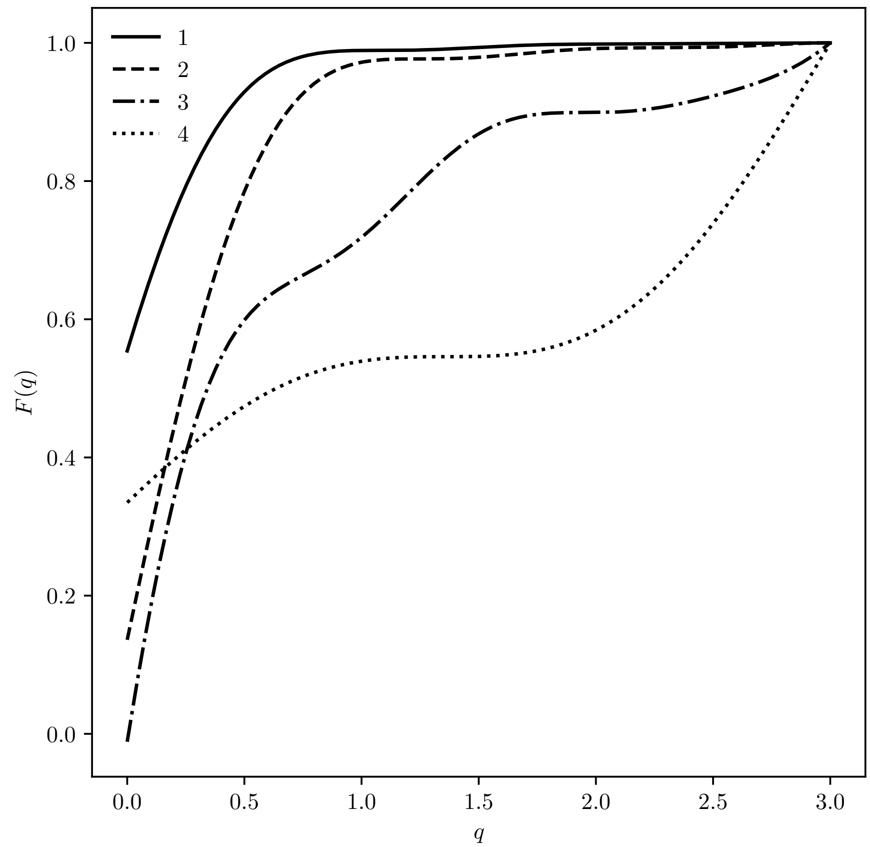
To explore more carefully the empirical content of the model, we first look carefully at four specific regimes which provide examples of what our method uncovers about liquidity, welfare, execution probabilities, and estimated beliefs about arrival rates. These regimes were chosen to have similar durations that are near the average regime duration. Summary statistics for these regimes are given in Table 4. For each regime, we report the three components of expected welfare per buyer: the friction component $\Gamma = 1 - F(1) + X_t - X_p$, the net difference in valuations Δ, and the probability of a buyer transacting, X_t. In addition, we report the separate components of Γ ($1 - F(1)$ and $X_t - X_p$), as well as $W/\Delta = \Gamma X_t$, which is the percent of the theoretical frictionless welfare gain Δ that is obtained given the frictions in the market.

Table 4: Descriptive statistics of selected regimes

	Spread	Welfare	Γ	Δ	X_t	$X_t - X_p$	$1 - F(1)$	$\frac{W}{\Delta}$
1	\$0.01	\$0.895	0.985	\$0.912	0.995916	0.974143	0.010965	98.1%
2	\$0.01	\$0.339	0.965	\$0.356	0.988683	0.936881	0.028103	95.4%
3	\$0.25	\$0.622	0.781	\$0.887	0.897984	0.499105	0.281690	70.1%
4	\$0.45	\$0.620	0.870	\$0.977	0.729066	0.409198	0.460588	63.4%

To begin, we compare regimes 1 and 2. These two regimes reflect a period of relative liquidity as indicated by a small spread, but they differ in some important ways. First, while both regimes have a competitive balance $1 - F(1)$ that strongly favors buyers, this is more than twice as strong in the first (0.011 vs. 0.028). Second, the welfare per buyer is significantly higher in regime 1 than in regime 2 (\$0.895 vs. \$0.339), but this is largely driven by a larger estimated difference in valuations of the asset (\$0.912 vs. \$0.356). The frictions of trading in a limit order market are estimated to be relatively low in these regimes as demonstrated by Γ close to 1. This comparison highlights the fact that liquidity and market efficiency are related, but separate concepts. Despite regimes 1 and 2 having the same spread, welfare in regime 2 is smaller than in regime 1 because differences in values between buyers and sellers is smaller, which means transactions do not realize as much welfare improvement. When welfare is normalized by the differences in valuations as in the last column of the table (W/Δ), these regimes are more similar (98.1% and 95.1%) — a consequence of both almost always having buyers transact (X_t) and having similar frictions (Γ). In Figure 3, the solid and dashed lines indicate the estimated arrival beliefs in the two regimes, where regime 2 is more likely to have more buyers arrive (indicated by a lower cumulative distribution at each q). This generates its higher competitive balance $1 - F(1)$.

Figure 3: Estimated cumulative distribution of order flows, $F(q)$, for selected regimes



Moving to regimes 3 and 4, both have larger spreads with a competitive balance tilting more towards sellers (especially in regime 4). Estimated welfare in both regimes 3 and 4 is higher than in regime 2, but lower than in regime 1. The estimated difference in valuations is largest of these 4 regimes in regime 4 (\$0.977). We notice that the friction multiplier Γ is smaller (0.781 and 0.870) for both of these regimes than regimes 1 and 2. As a fraction of potential frictionless gains, regime 3 achieves 70.1% while regime 4 achieves only 63.4%. These two regimes have similar expected welfare per buyer, but different components drive this welfare. For regime 3, market frictions are significant so the resulting Γ is lower, while the probability of buyers transacting (X_t) is higher. Meanwhile the market frictions for regime 4 weigh less on expected welfare, but the probability of buyers transacting is lower. In Figure 3, the dot-dashed and dotted lines show that buyers in these two regimes are likely to be more plentiful, tilting the competitive balance toward sellers and making buyers less likely to transact.

These regime comparisons highlight an important takeaway from this paper, which is that liquidity is not necessarily a proxy for the overall welfare of market participants. Regime 2 has the lowest estimated welfare gain of the four regimes despite having a spread of \$0.01. This lower welfare is driven by relatively low valuation differences. However, welfare for regimes 3 and 4 is less than for regime 1 because market frictions associated with participating in the limit order market are larger in these regimes.

3.4 The distributions of Δ, Γ, Welfare and related values

Moving beyond these case studies, we now study the entire sample of estimated regime characteristics. We report the distribution of estimates for welfare, valuation differences Δ, aggregate frictions Γ, the probability of buyer's transacting X_t, the competitive balance $1-F(1)$, buyers' welfare multiplier $X_t - X_p$, and the fraction of potential welfare achieved in Table 5. First, we note that the distribution of valuation differences Δ is highly skewed, with a mean of \$0.989 but a median of \$0.454. Both of these valuation differences are relatively small, considering the average asset price is \$28,219 in this sample.

Table 5: Distributions of competitive balance, market welfare, and its components

	Spread	Welfare	Δ	Γ	X_t	$1-F(1)$	$X_t - X_p$	W/Δ
Mean	\$0.16	\$0.819	\$0.989	0.927	0.947	0.124	0.803	88.1%
Coeff. of Var.	\$2.35	\$1.267	\$1.316	0.068	0.093	1.497	0.286	0.1%
25%	\$0.01	\$0.241	\$0.260	0.896	0.946	0.023	0.739	83.4%
50%	\$0.01	\$0.422	\$0.454	0.956	0.985	0.039	0.915	94.0%
75%	\$0.08	\$0.857	\$0.985	0.971	0.991	0.139	0.948	96.2%

The distribution of $1-F(1)$ shows that market makers were quite competitive throughout the sample period, with a median value of 0.039 and a mean of 0.124. Of the components that make up welfare, the coefficient of variation indicates that X_t and Γ are also less volatile than Δ. As a result, the fraction of potential welfare that are realized (W/Δ) is fairly stable, with a median of 94.0% and a mean of 88.1%. Figure 4 plots the distribution of the components of welfare, each of which are quite skewed.

3.5 Understanding the determinants of liquidity

This section explores the determinants of the spread and price impact over the regimes studied here. Equation (16) decomposes the spread into the valuation differences between buyers and sellers (Δ) and the measure of seller competitiveness ($1 - F(1)$). Table 6 gives the correlation between the log spread, and logged values of $1 - F(1)$ and Δ. Logs are taken

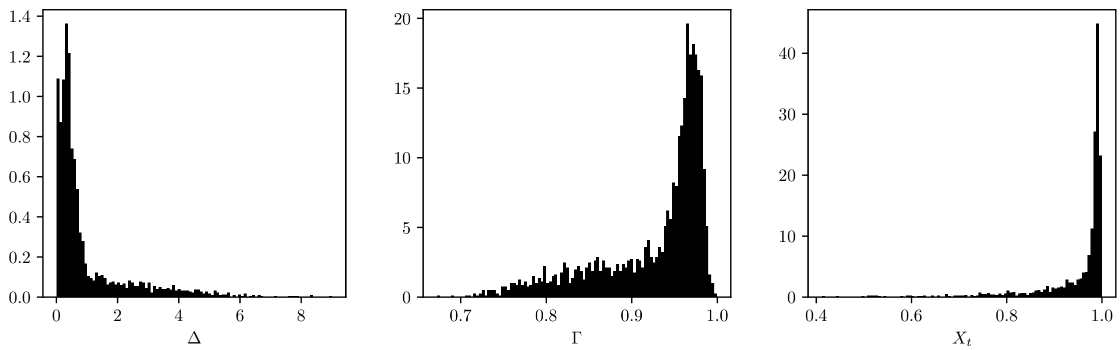


Figure 4: Histograms of the components of welfare

because equation (16) suggests that the relationship between these logged values is linear. We see that $\log(\Delta)$ is more highly correlated with spread than $\log(1 - F(1))$. In the estimation process, we use the shape of the order book to estimate $F(1)$ and directly observe the spread. Values for Δ are then estimated as a residual. Since the data-generated predictions of the spread from the model arise out of aggregating observations into the estimate of competitive balance $1 - F(1)$, we regress $\log(S)$ on $\log(1 - F(1))$ to see how much of the total spread is predicted by this data-generated market statistic. We find that this regression yields an R^2 of 0.30. Thus, the competitive balance as measured by $1 - F(1)$ captures about 30% of the variation in the observed spread for this sample.

Table 6: Correlation between determinants of liquidity

	$\log(S)$	$\log(1 - F(1))$	$\log(\Delta)$
$\log(S)$	1.000		
$\log(1 - F(1))$	0.657	1.000	
$\log(\Delta)$	0.706	-0.070	1.000

To understand the role of competitive balance in the spread, Figure 5 shows how the distribution of spread would change if Δ was assumed to stay as it is in each regime while the competitive balance is changed from its estimated values to $1 - F(1) = 0.5$ (for each regime). As can be seen graphically and in Table 7, the mean spread would increase from \$0.159 to \$0.495, and the standard deviation would increase from \$0.374 to \$0.651. The median spread in this world would be more than 22 times higher, going from \$0.01 to \$0.227. This suggests that while estimated valuation differences are very important in understanding the spread over this sample, the competitive balance component of the spread is also an important phenomenon to understand.

With the structure of our model, we can also measure expected welfare that accrue to

Figure 5: The predicted distribution of spread if competitive balance were $1 - F(1) = 0.5$

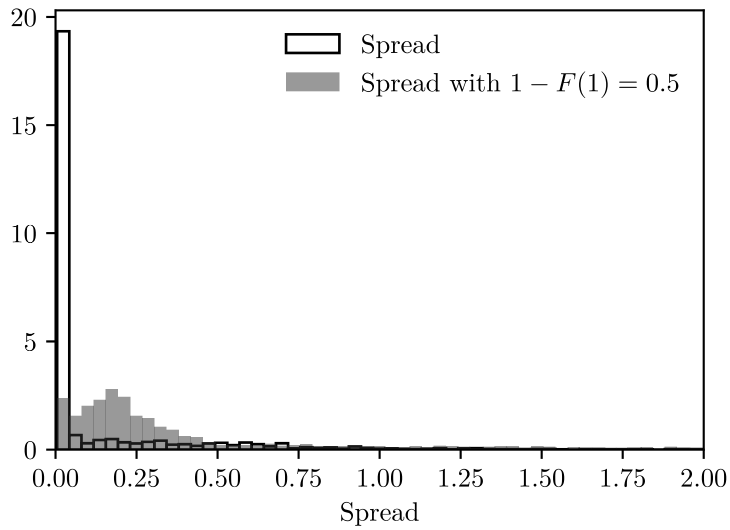


Table 7: The distribution of realized spread and counterfactual spread if competitive balance were $1 - F(1) = 0.5$

	Realized Spread	Counterfactual Spread
Mean	\$0.159	\$0.495
St. Dev.	\$0.374	\$0.651
Min	\$0.010	\$0.005
25%	\$0.010	\$0.130
50%	\$0.010	\$0.227
75%	\$0.080	\$0.493
Max	\$4.230	\$4.505

sellers and to buyers. Here we plot those on a per-regime basis (as opposed to per buyer basis). From equation (11), we see that the seller's utility gain is $\Delta(1 - F(1))$ and from equation (12), the buyer's utility gain is $\Delta(X_t - X_p)$. Figure 6 shows the distribution of these two components of welfare (left) and the fraction of the total gains that accrue to sellers (right). Table 8 gives summary statistics of the distribution of these two components of welfare. We can see that buyers' estimated welfare gain is significantly higher and more diffuse across regimes than sellers' welfare gain.

3.6 Robustness

We now consider how our welfare estimates are impacted by alternative assumptions used in estimating the distribution of arrivals, $F(q)$. We focus on two free parameters: the asset

Table 8: The distribution of expected welfare for buyers and sellers across regimes

	Buyer Welfare	Seller Welfare	Fraction of Welfare to Sellers
Mean	\$0.728	\$0.155	13.969%
Coeff. of Var.	\$1.297	\$2.340	1.496%
Min	\$0.000	\$0.010	0.205%
25%	\$0.215	\$0.010	2.382%
50%	\$0.396	\$0.010	4.119%
75%	\$0.761	\$0.080	15.895%
Max	\$8.820	\$4.230	99.967%

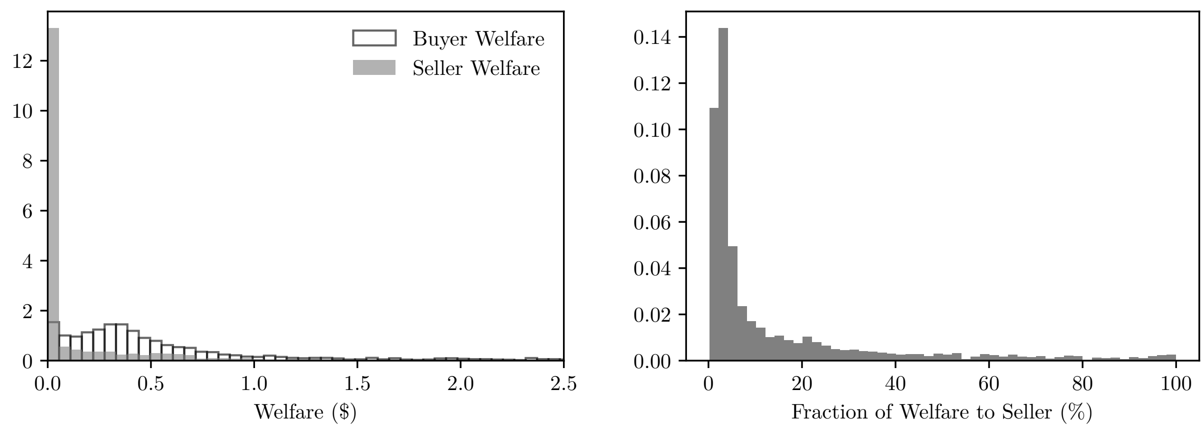


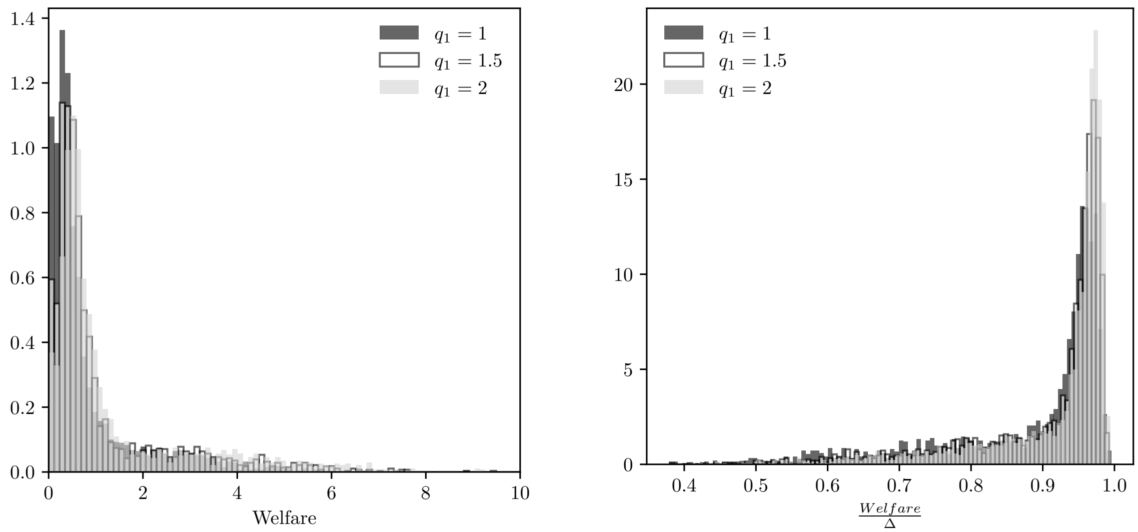
Figure 6: The distribution of seller and buyer welfare (left) and the fraction of total welfare accruing to sellers (right)

quantity corresponding to $q = 1$, and the upper limit of the support of q.

In the previous sections we set $q = 1$ to correspond to the execution of 1 BTC, which was nearly the maximum of observed execution sizes. Even so, here we repeat our analysis under alternatives where $q = 1$ is instead ascribed to the execution of 1.5 or 2 BTC. The resulting distribution of welfare across regimes is compared in Figure 7, while summary statistics of this distribution are reported in Table 9. As we raise the executions size for $q = 1$, we see that welfare estimates increase somewhat — the mean welfare increases by \$0.549 moving $q = 1$ from 1 to 2 BTC. Nearly all of this increase comes from a larger estimated Δ, which can be seen by looking to W/Δ. The realized percentage of potential gains rises only four percentage points, and the median rises by two percentage points.

In Section 3.2, we also truncated the order book with $q_{max} = 3$, meaning that we ignored the presence of limit orders that would have required market orders totalling more than 3 bitcoins to transact in rapid succession, which never occurs in our data. Here, we evaluate

Figure 7: The distribution of welfare (left) and $\frac{W}{\Delta}$ (right) for $q = 1$ corresponding to BTC values of 1, 1.5 and 2



welfare under alternative assumptions of $q_{max} = 2$ or $q_{max} = 4$, while assuming that $q = 1$ still represents transactions totaling 1 BTC. Figure 8 compares the distribution of welfare under each assumption, while Table 10 gives summary statistics for these distributions. This indicates that the estimated welfare falls as q_{max} increases. The estimates have increased noise with a small q_{max}, as seen in the coefficient of variation. Again, this is mostly driven by changes in Δ, since the distribution of W/Δ has much less movement.

Taken together, these robustness results suggest that the model is qualitatively stable to changes in the parameterizations considered here, particularly in estimating the realized welfare as a percentage of potential gains.

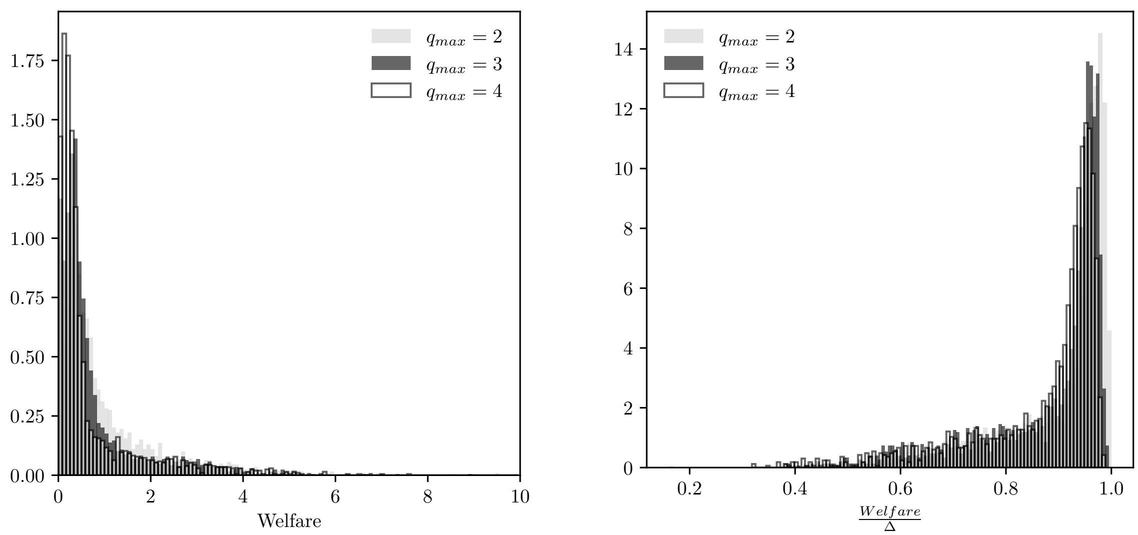
4 Conclusion

Limit order book markets create different roles for liquidity providers and demanders, where the former set a price and await a transaction, while the latter show up and take the best available price. Our model is built on this essential feature. Both sets of parties anticipate the arrival of market participants to predict the price at which transactions will occur. Thus, the observed order book and its inherent liquidity, including spread and price impact, are an equilibrium consequence of beliefs about the relative size of liquidity suppliers versus that of liquidity demanders. The liquidity implications of this distribution of buyer arrival can be summarized in the competitive balance, which determines how the gains from trade are split between buyers and sellers. We demonstrate that as the competitive balance favors sellers,

Table 9: The distribution of welfare (left) and $\frac{W}{\Delta}$ for $q = 1$ corresponding to BTC values of 1,1.5 and 2

	Welfare			$\frac{W}{\Delta}$		
	$q_1 = 1$	$q_1 = 1.5$	$q_1 = 2$	$q_1 = 1$	$q_1 = 1.5$	$q_1 = 2$
Mean	\$0.819	\$1.112	\$1.368	88.10%	90.65%	92.14%
Coeff. of Var.	\$1.267	\$1.234	\$1.186	0.14%	0.11%	0.10%
Min	\$0.004	\$0.005	\$0.004	37.85%	38.04%	41.98%
25%	\$0.241	\$0.342	\$0.447	83.43%	88.44%	91.29%
50%	\$0.422	\$0.562	\$0.680	94.04%	95.41%	96.24%
75%	\$0.857	\$1.111	\$1.412	96.20%	97.00%	97.45%
Max	\$8.880	\$10.461	\$13.719	99.55%	99.24%	99.39%

Figure 8: The distribution of welfare (left) and $\frac{W}{\Delta}$ (right) for $q_{max} \in \{2, 3, 4\}$



market liquidity falls. Even so, less liquidity can generate greater welfare when it means more parity in the flow of buyers and sellers, leading to a greater volume of trades.

Using data from the Coinbase BTC/USD exchange, we demonstrate how order book data can be interpreted in the light of our model. We identify time periods (regimes) during which the market plausibly appears stable enough to apply the model. For this market, we estimate that welfare is 88% of what a frictionless market could generate (on average), per buyer. Across the regimes in our data, we find that the variation in welfare is largely driven by variation in valuation differences, but that the frictions in the market also play an important role in reducing welfare.

Further work in this area could investigate the competitive balance across alternative exchanges of the same asset. This work would allow us to characterize the pricing and

Table 10: The distribution of welfare (left) and $\frac{W}{\Delta}$ for $q_{max} \in \{2,3,4\}$

	Welfare			$\frac{W}{\Delta}$		
	$q_{max}=2$	$q_{max}=3$	$q_{max}=4$	$q_{max}=2$	$q_{max}=3$	$q_{max}=4$
Mean	\$1.791	\$0.826	\$0.690	57.2%	55.0%	54.4%
Coeff. of Var.	2.527	1.644	1.773	0.1%	0.1%	0.1%
Min	\$0.000	\$0.000	\$0.001	0.0%	0.8%	15.4%
25%	\$0.221	\$0.155	\$0.101	57.5%	55.2%	54.3%
50%	\$0.414	\$0.273	\$0.196	58.9%	58.0%	57.2%
75%	\$1.373	\$0.631	\$0.474	59.3%	58.7%	58.2%
Max	\$59.418	\$8.818	\$8.209	59.9%	59.7%	59.6%

welfare efficiency across exchanges. Such an effort could yield insights into how exchanges compete, the welfare and/or losses associated with this competition, as well as the role of market regulation in promoting such competition. This model also allows for the study of changes in fee structures or other competitive factors across regimes and their potential welfare and liquidity effects.

A Proof of Proposition 1

Proof. First, we verify that the proposal satisfies the equilibrium conditions.

Note that by substituting Eq. 7 into Eq. 8, we readily obtain $G(a(q)) = 1 - F(q)$, satisfying condition 3.

Moreover, since $F'(q) > 0$ for $q \in [0, 1]$, the support of G will run from $a(0)$ (as F is not defined for $q < 0$) to $a(1)$ (as G is not defined for $a > a(1)$), fulfilling condition 1.

To verify condition 4, we substitute $a(s)$ from Eq. 7 into Eq. 3:

$$V_B = \int_0^1 \left(\int_0^q \left(1 - \frac{1 - F(1)}{1 - F(s)} \right) ds \right) \frac{\Delta F'(q)}{q} dq + \int_1^\infty \left(\int_0^1 \left(1 - \frac{1 - F(1)}{1 - F(s)} \right) ds \right) \frac{\Delta F'(q)}{q} dq.$$

After moving out terms that are constant w.r.t. integration, this simplifies to:

$$\begin{aligned} V_B &= \Delta \Bigg(F(1) - (1 - F(1)) \int_0^1 \left(\int_0^q \frac{1}{1 - F(s)} ds \right) \frac{F'(q)}{q} dq \\ &\quad + \int_1^\infty \frac{F'(q)}{q} dq - (1 - F(1)) \int_1^\infty \left(\int_0^1 \frac{1}{1 - F(s)} ds \right) \frac{F'(q)}{q} dq \Bigg). \end{aligned}$$

Changing the order of integration allows the last two double integrals to be combined:

$$V_B = \Delta \left(F(1) + \int_1^\infty \frac{F'(q)}{q} dq - (1 - F(1)) \int_0^1 \left(\int_s^\infty \frac{F'(q)}{q} dq \right) \frac{1}{1 - F(s)} ds \right).$$

Utilizing the notation for X_t and X_p, we obtain:

$$V_B = \Delta \left(X_t - X_p \right). \tag{21}$$

Moreover, buyers indifferent about completing the transaction at any of the offered prices, since $\bar{a} = d_B - f_t$.

Finally, for condition 1, consider any price a in the support of $G(a)$. After substituting $G(a)$ using Eq. 8, we obtain:

$$V_S(a) = d_A + \Delta \cdot (1 - F(1)).$$

This is constant w.r.t. a, so all prices in the support are equally profitable.

If a firm offered a price below $a(0)$, they would transact with the same probability (that is, with probability 1) as when offering $a(0)$, but with a strictly lower price and thus less profit. If a firm offered a price above $a(1)$, it will always be rejected since it is above the reservation price. Thus, everything in the support is profit maximizing and nothing outside of it is.

To see that this is the unique equilibrium, we first prove that there cannot be an equilibrium solution that contains atoms. Suppose at price $\hat{a} \geq a(0)$ there is an atom of weight $g(\hat{a}) > 0$. A seller who places a limit order of $\hat{a}$ will transact with probability $G(\hat{a})$, after all other sellers with orders at or below $\hat{a}$, earning expected profit:

$$E\pi(\hat{a}) = G(\hat{a})(\hat{a} + f_m) + (1 - G(\hat{a}))d_A. \tag{22}$$

If the seller instead places the limit order at $\hat{a} - \epsilon$ for some $\epsilon > 0$, then the probability of transacting would be $G(\hat{a} - \epsilon)$ and the expected profit would be

$$E\pi(\hat{a} - \epsilon) = G(\hat{a} - \epsilon)(\hat{a} - \epsilon + f_m) + (1 - G(\hat{a} - \epsilon))d_A. \tag{23}$$

Note that $\lim_{\epsilon \searrow 0} G(\hat{a} - \epsilon) = G(\hat{a}) + g(\hat{a})$, so that the seller is jumping ahead of the mass of sellers with orders at $\hat{a}$. Thus, there exists a sufficiently small $\epsilon > 0$ for which $E\pi(\hat{a} - \epsilon) > E\pi(\hat{a})$. Thus, the expected profit from placing an order at $\hat{a} - \epsilon$ is strictly greater than the expected profit from placing an order at $\hat{a}$ which violates the equal profit condition and hence is not an equilibrium.

For atom-less distributions over $G(a)$, the interior of the support is unique because of the equal profit condition for sellers. The equal profit condition when applied to Eq. 1 implies a differential equation for $G(a)$,

$$G'(a)\,(a + f_m - d_A) + G(a) = 0. \tag{24}$$

This equation has a unique solution.[20] Thus, $G(a)$ is uniquely defined. Then, the equilibrium requirement that $G(a)$ be consistent with the distribution $F(q)$ given in Eq. (2) uniquely defines $a(q)$. In this relationship, since $F(q)$ is assumed to be strictly monotonic and $G(a)$ is shown to be also monotonic, $a(q)$ is unique. Last, we note that the endpoints of the support of a are uniquely defined. □

[20]The general solution for a first-order differential equation of this form is

$$G(a) = -\frac{c_1}{a + f_m - d_A}, \tag{25}$$

while the constant c_1 is pinned down by the necessity that $G(a(0)) = 1$.

B Proof of Proposition 2

Proof. The results for V_S follow immediately from Eq. 11. The change in V_B move in direct proportion to $X_t - X_p$.

Part 1: If we let $\chi \equiv \int_1^\infty \frac{1}{q} F'(q)dq$, we can rearrange to get:

$$\begin{aligned}
X_t - X_p &= F(1) + \int_1^\infty \frac{1}{q} F'(q)dq - (1 - F(1)) \int_0^1 \left(\int_s^\infty \frac{F'(q)}{q} dq \right) \frac{1}{1 - F(s)} ds \\
&= F(1) + \chi - (1 - F(1)) \int_0^1 \left(\int_s^1 \frac{F'(q)}{q} dq + \chi \right) \frac{1}{1 - F(s)} ds \\
&= F(1) + \chi \left(1 - \int_0^1 \frac{1 - F(1)}{1 - F(s)} ds \right) - (1 - F(1)) \int_0^1 \left(\int_s^1 \frac{F'(q)}{q} dq \right) \frac{1}{1 - F(s)} ds.
\end{aligned}$$

Note that $\chi \equiv \int_1^\infty \frac{1}{q} F'(q)dq$ is larger under $\hat{F}$ as it places more weight where $1/q$ is bigger. All other terms will be the same under F or $\hat{F}$ since they only involve $q \leq 1$. Finally, the first parenthetical element is positive because $F(1) > F(s)$ for $s < 1$.

Part 2: Using the same rearrangement, in the second line, χ is the same under F or $\hat{F}$. Moreover, $\frac{1}{1-F(s)} > \frac{1}{1-\hat{F}(s)}$, which makes each of the integrals smaller under $\hat{F}$. The second integral w.r.t. q is also smaller for each q, and furthermore $\hat{F}'$ will place more weight where there is a smaller $1/q$. Thus the double integrals are smaller, and with the negative, the whole term is larger under $\hat{F}$. □

C Proof of Proposition 3

Proof. By applying Proposition 2, we find that $\hat{X}_t - \hat{X}_p$ is larger than $X_t - X_p$ in either case. Meanwhile, $X_t = F(1) + \chi$ is also larger under $\hat{F}(q)$ in the first scenario or constant in the second. Thus, their product is unambiguously larger under $\hat{F}(q)$. □

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